Probability theory and statistics

Lecture Notes

P&S team

Ukrainian Catholic University

3rd term
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APPLIED SCIENCES FACULTY

Lecture 4. Discrete random variables

Outline

1. RV's: Definitions, examples, and properties
   - Examples
   - Definition
   - Probability mass function
   - Cumulative distribution function
2. Standard discrete distributions
   - Binomial distribution
   - Geometric distribution
   - Poisson distribution
3. Joint distribution and independence
   - Examples
   - Definitions
   - Independence
   - Conditional distribution

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Random variables: Examples

Often, outcomes are associated with some numerical values:

- In rolling two fair dice, only the total sum X can be important; e.g., $X = 5$ for outcomes $(1,4), (2,3), (3,2), (4,1)$; therefore, $\mathrm{P}(X = 5) = 4/36$. X can assume values between 2 and 12:

Value x of X	event $\{X = x\}$	$\mathrm{P}(X = x)$
2	(1,1)	1/36
3	(1,2),(2,1)	2/36
...	...	...
12	(6,6)	1/36

Random variables: Examples

- In tossing a biased coin three times, record the total number Y of Heads; thus Y can assume the values $0, 1, 2, 3$:

Value y of Y	event $\{Y = y\}$	$\mathrm{P}(Y = y)$
0	'TTT'	q^3
1	'HTT', 'THT', 'TTH'	$3pq^2$
2	'HHT', 'HTH', 'THH'	$3p^2q$
3	'HHH'	p^3

Random variable: Example

- Shuffle n cards, and let X be the position of the jth card.
- $\mathrm{P}(X = k) = \frac{1}{n}$

- Your example

Random variable: Example

3 balls are drawn without replacement from an urn containing 10 balls numbered 1 to 10. What is the probability that the largest no. is k?

- Sample space

$$\Omega = \{\textit{"all draws"}\} = \{\textit{subsets of 3 from } 1, \ldots, 20\}$$
$$= \{\omega = \{\omega_1, \omega_2, \omega_3\} \mid \omega_1 < \omega_2 < \omega_3, \omega_j \in \overline{1, 10}\}$$

- $\mathscr{F} = 2^{\Omega}$; P is a uniform probability measure on Ω
- $X(\omega) =$"*largest of three numbers*" $= \omega_3$
- $X(\omega) = k \iff$ ball no. k is drawn and the other two are smaller:

$$\iff \omega_1 < \omega_2 < \omega_3 = k$$

- therefore, $\mathrm{P}(X = k) = C_{k-1}^2 / C_{10}^3$

k	3	4	5	6	7	8	9	10
$\mathrm{P}(X = k)$								

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Definition of a random variable

Definition

A random variable (**rv**) ξ on a probability space $(\Omega, \mathscr{F}, \mathsf{P})$ is any[a] function $\xi : \Omega \to \mathbb{R}$

[a] ξ must be *measurable*, ie., sets like $\{\xi \leq t\}$ should be events (from $\mathscr{F}$)

Example

The *lifetime* T of any equipment (eg., notebook hard disk) can be any positive number. Here

- the sample space $\Omega = \{\text{the set of all notebooks}\}$;
- probability measure P on Ω — any

T is a random variable on this probability space.

Definition

A random variable X is discrete if it assumes only finite or countable infinite set of values $x_1, x_2, \ldots$

Definition of a random variable

Notation

X random variable x numerical value

Example of a random variable

3 balls are drawn without replacement from an urn containing 10 balls numbered 1 to 10. What is the probability that the largest of three numbers will be at least 7?

- Sample space $\Omega = \{\textit{"all draws"}\} = \{\textit{subsets of 3 from } 1, \ldots, 20\}$
- $\mathscr{F} = 2^{\Omega}$; P a uniform probability measure on Ω
- $X =$"*largest of three numbers*" is a **rv** on $(\Omega, \mathscr{F}, \mathrm{P})$
- $X = k \iff$ ball no. k is drawn and the other two are smaller:

$$\mathrm{P}(X = k) = \frac{\binom{k-1}{2}}{\binom{10}{3}}$$

- in particular,

$$\begin{aligned} &\mathrm{P}(X \geq 7) \\ &\quad = \mathrm{P}(X = 7) + \mathrm{P}(X = 8) + \mathrm{P}(X = 9) + \mathrm{P}(X = 10) \\ &\quad = 5/6 \end{aligned}$$

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Probability mass function

In the previous example, constitution of the event $\{X = k\}$ was not of considerable interest; only the probability $\mathrm{P}(X = k)$ was important

Definition (Probability mass function)

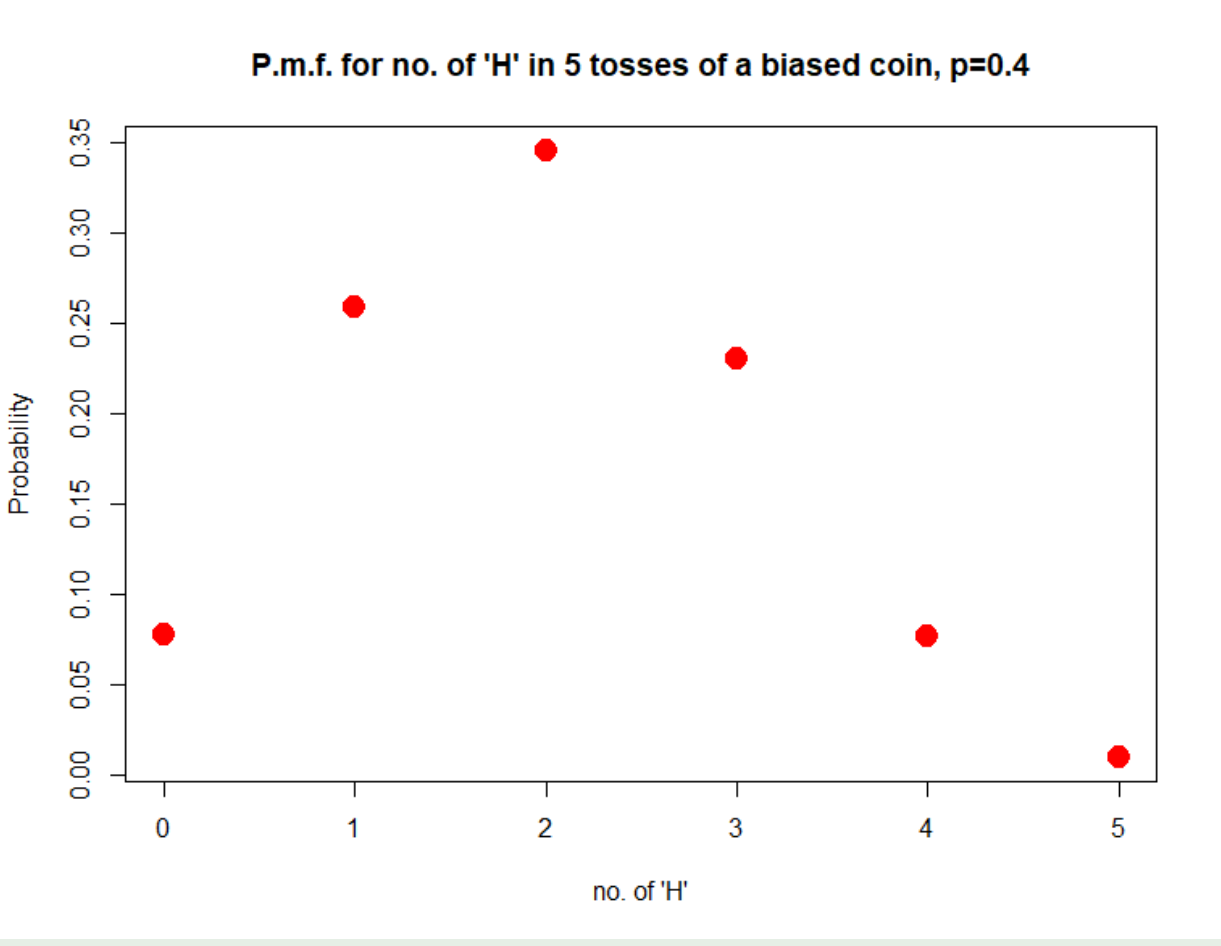
For a discrete **r.v.** X its probability mass function (**pmf**) p_X is given by

$$p_X(a) = \mathrm{P}(X = a), \qquad a \in \mathbb{R}.$$

- If X assumes values $x_1, x_2, \ldots$, then $p_X(a) \geq 0$ for $a = x_k$ and $p_X(a) = 0$ otherwise;
- $\sum_{k \geq 0} p_X(x_k) = 1$ since $\{X = x_k\}_{k>0}$ form a partition of Ω

Probability mass function

Example (Pmf for no. of 'H' in 5 tosses of a biased coin ($p = .4$))



Random variables with the same **pmf**

Warning: different **rv** may posses the same **pmf**

- total number X of `Heads` in 5 tosses of biased coin with $p = .4$:

$$p_X(k) = \mathrm{P}(X = k) = \binom{5}{k} p^k q^{5-k}, \qquad k = 0, 1, \ldots, 5$$

- an urn contains 4 white balls and 6 black ones. A ball is drawn at random, its colour marked, and then returned in the urn. The number Y of white balls in 5 draws is a **rv** with

$$p_Y(k) = \mathrm{P}(Y = k) = \binom{5}{k} \left(\tfrac{4}{10}\right)^k \left(\tfrac{6}{10}\right)^{5-k}, \qquad k = 0, 1, \ldots, 5$$

- generate 5 random numbers from $[0, 1]$ and count the no. Z of those $\leq .4$; then

$$\mathrm{P}(Z = k) = \mathrm{P}(Y = k) = \mathrm{P}(X = k)$$

Random variables with the same **pmf**

Remark (on **pmf**)

*For many purposes, it does not matter how a **rv** X was defined;*
*it suffices to know its **pmf** p_X:*
*then we know the **values** of X and the **probabilities***

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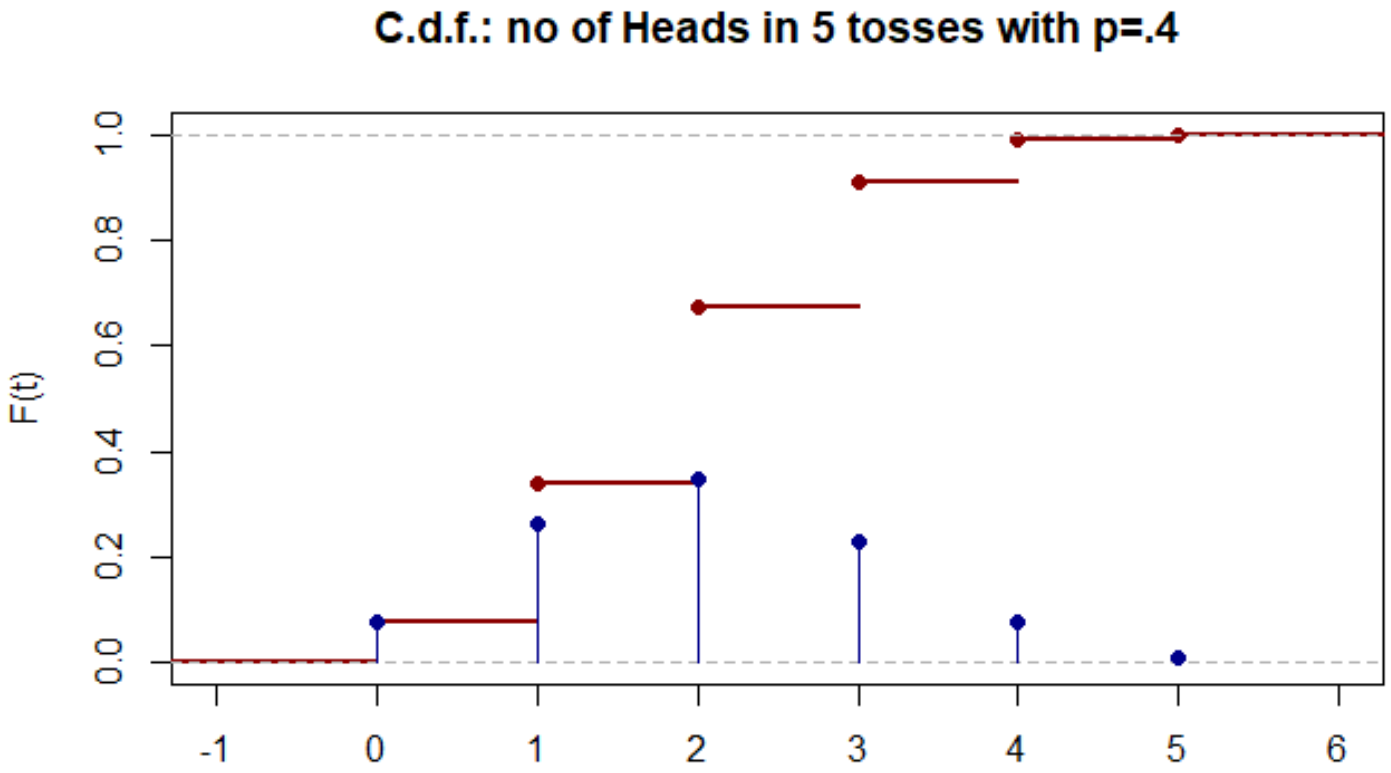
Cumulative distribution function

Alternatively, a **r.v.** can be characterised by its **cdf**:

Definition (cdf)

The cumulative distribution function F_X of a **rv** X is a function on $\mathbb{R}$ given by

$$F_X(t) := \mathrm{P}(X \leq t)$$



Cumulative distribution function (cont)

Properties of **cdf**:

- $0 \leq F_X(t) \leq 1$;
- F_X is non-decreasing, with $\lim_{t\to-\infty} F_X(t) = 0$, $\lim_{t\to+\infty} F_X(t) = 1$
- F_X is right-continuous: $F_X(t) = F_X(t+0) := \lim_{\varepsilon\to 0} F_X(t+\varepsilon)$;

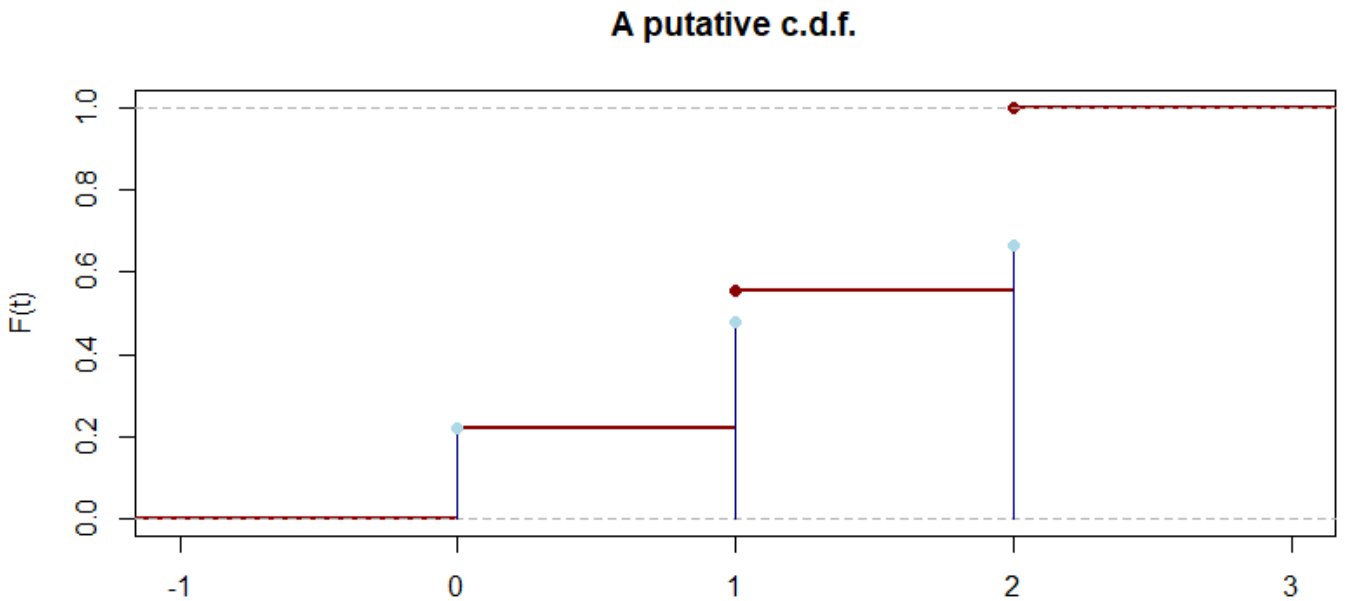
  however, $F_X(t-0) := \lim_{\varepsilon\to 0} F_X(t-\varepsilon)$ need not be equal to $F_X(t)$

pmf vs **cdf**:

- Clearly, $F_X(t) = \sum_{x_j \leq t} p_X(x_j)$
- on the other hand, $p_X(x_j) = F_X(x_j) - F_X(x_j - 0)$

Question

- *Why is distribution called distribution?*
- *Given some putative **cdf** F, do we know there is $(\Omega, \mathscr{F}, \mathrm{P})$ and X s.t. $F_X = F$?*



Definition

We say a **rv** X has distribution F (written $X \sim F$) if $F_X = F$

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Bernoulli distribution $\mathscr{B}(1, p)$

- Assume a given experiment can result in either
  - *success* with probability $p \in (0, 1)$, or
  - *failure* with probability $1 - p$
- Let event A denote the "success"
- its indicator χ_A is

$$\chi_A(\omega) = \begin{cases} 1 & \omega \in A; \\ 0 & \omega \notin A \end{cases}$$

- the indicator of success χ_A is a Bernoulli **rv** (written $\chi_A \sim \mathscr{B}(1, p)$)

Example (Bernoulli rv's:)

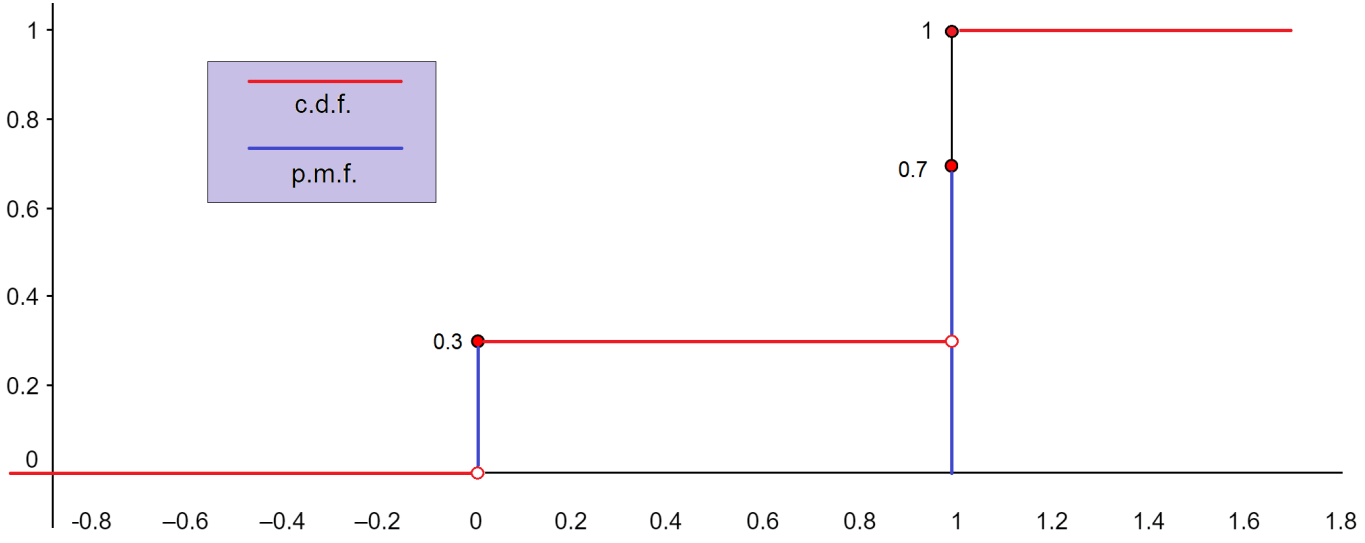
- indicator of getting at least two *Heads* in tossing a coin 3 times
- indicator of getting *six* in rolling a die
- indicator of getting *nine* in rolling a die twice

Bernoulli distribution $\mathscr{B}(1, p)$

If $X \sim \mathscr{B}(1, p)$ is a Bernoulli **rv**, then its **pmf** is as follows:

$$p_X(x) = \begin{cases} p, & \text{if} \quad x = 1; \\ 1 - p, & \text{if} \quad x = 0 \end{cases}$$

Vice versa, any **rv** Y assuming values 0 and 1 only is Bernoulli:
with $A := \{\omega \mid Y(\omega) = 1\} = \{Y = 1\}$, $Y = \chi_A$



Binomial distribution $\mathscr{B}(n, p)$

If a Bernoulli experiment is repeated n times, and the outcomes of the different experiments are independent, then the total number of successes is a binomial **rv** $X \sim \mathscr{B}(n, p)$:

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

By binomial thm,

$$\sum_{k=0}^{n} p_X(k) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = (p + 1 - p)^n = 1$$

Example (Number of sixes in rolling a die 5 times:)

If Y is the number of sixes, then

$$p_Y(k) = \binom{5}{k} \left(\tfrac{1}{6}\right)^k \left(\tfrac{5}{6}\right)^{5-k}$$

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Geometric **rv** $\mathscr{G}(p)$

The number of Bernoulli experiments until the first success is a **rv** with geometric distribution:

$$p_X(k) = p(1-p)^{k-1}, \qquad k = 1, 2, \ldots$$

Observe that (and that is Murphy's law)

$$\sum_{k\geq 1} p_X(k) = p\sum_{j=0}^{\infty}(1-p)^j = \frac{p}{1-(1-p)} = 1$$

Example (Number of die throws until the first six)

If Y is the number of rolling until the first six shows up, then

$$p_Y(k) = \left(\tfrac{1}{6}\right)\left(\tfrac{5}{6}\right)^{k-1}$$

Memorylessness

Memoryless rv

We say that a nonnegative random variable X is *memoryless* if

$$\mathrm{P}\{X > s + t | X > t\} = \mathrm{P}\{X > s\} \text{ for all } s, t \geq 0$$

A geometrically distributed random variable X is memoryless.

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Poisson distribution $\mathscr{P}(\lambda)$

The Poisson **rv** $X \sim \mathscr{P}(\lambda)$ is the limit of binomial **rv**'s $X_n \sim \mathscr{B}(n, p_n)$ when $n \cdot p_n \to \lambda$ as $n \to \infty$; then

$$p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \qquad k = 0, 1, \ldots$$

Again,

$$\sum_{k \geq 0} p_X(k) = e^{-\lambda} \sum_{k \geq 0} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1$$

Example (Radioactive decay: alpha-particle emission)

Assume the number of alpha-particles emitted by some radioactive substance in one hour is distributed according to the Poisson law with parameter 6. Find the probability that at least 3 alpha-particles will be emitted within the next hour.

Since $\{X \geq 3\} = \Omega \setminus \{X \leq 2\}$, we find that

$$\mathrm{P}(X \geq 3) = 1 - \mathrm{P}(X \leq 2) = 1 - \left(1 + \lambda + \tfrac{\lambda^2}{2}\right) e^{-\lambda}$$

Poisson distribution as a limit of binomial ones

Alpha-particle emission in shorter times

- Denote by $Y_1, Y_2, \ldots, Y_{60}$ the number alpha-particle emissions during the 1st, 2nd etc minute
- We expect that $Y_k \sim \mathscr{B}(1, 0.1)$; thus $Y := Y_1 + \cdots + Y_{60} \sim \mathscr{B}(60, 0.1)$
- Let us compare the distributions of $X \sim \mathscr{P}(6)$ and $Y \sim \mathscr{B}(n, 6/n)$ for various n

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An example of joint distribution

Example

An urn contains 3 white balls and 2 black ones. Three balls are drawn from the urn in turn without replacement. Set

- X is the total number of white balls drawn
- Y is the number of white balls in the first two draws

	WWW	WWB	WBW	BWW	WBB	BWB	BBW
P	.1	.2	.2	.2	.1	.1	.1
X	3	2	2	2	1	1	1
Y	2	2	1	1	1	1	0

The joint distribution of (X, Y) can be given by the table

	X=1	X=2	X=3
Y=0	.1	0	0
Y=1	.2	.4	0
Y=2	0	.2	.1

Marginal distributions of X and Y

The joint distribution completely describes the **pmf**'s of X and Y, eg:

$$p_X(1) = \mathrm{P}(X = 1) = \mathrm{P}(X = 1, Y = 0) + \mathrm{P}(X = 1, Y = 1) + \mathrm{P}(X = 1, Y = 2) = .3;$$

$$p_Y(1) = \mathrm{P}(Y = 1) = \mathrm{P}(X = 1, Y = 1) + \mathrm{P}(X = 2, Y = 1) + \mathrm{P}(X = 3, Y = 1) = .6$$

	X=1	X=2	X=3
Y=0	.1	0	0
Y=1	.2	.4	0
Y=2	0	.2	.1

	X=1	X=2	X=3
Y=0	.1	0	0
Y=1	.2	.4	0
Y=2	0	.2	.1

Conditional distributions of X and Y

If the value of Y is known, this changes the possibilities for X, e.g.

$$\begin{aligned}
\mathrm{P}(X = 1 \mid Y = 0) &= \mathrm{P}(\mathrm{BBW} \mid \mathrm{BBW}) = 1, \\
\mathrm{P}(X = 2 \mid Y = 0) &= = 0, \\
\mathrm{P}(X = 3 \mid Y = 0) &= = 0,
\end{aligned}$$

and this gives a conditional distribution of X given the event $\{Y = 0\}$.

Definition

The function $p_{X|Y}(x \mid y) := \mathrm{P}(X = x \mid Y = y)$ for a fixed y is the **pmf** of a random variable $X \mid Y = y$ that is said to have a conditional distribution given that $\{Y = y\}$

Observe that $X \mid Y = y$ assumes the same values as X but their **pmf** are in general different

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Joint probability mass function

Assume X and Y are two **rv** on a probability space $(\Omega, \mathscr{F}, \mathsf{P})$

Then we have a random mapping

$$\Omega \ni \omega \mapsto \big(X(\omega), Y(\omega)\big) \in \mathbb{R}^2$$

Definition (Joint probability mass function)

Assume X and Y are discrete **rv** on $(\Omega, \mathscr{F}, \mathsf{P})$; then the function

$$p_{X,Y}(x, y) := \mathsf{P}(X = x, Y = y)$$

is called the joint probability mass function

Remark (Properties of joint **pmf**)

*$p_{X,Y}$ satisfies all properties of the usual **pmf**, namely*

$$p_{X,Y} \geq 0 \qquad \textit{and} \qquad \sum_{x,y} p_{X,Y}(x, y) = 1$$

Marginal **pmf's** of X and Y in a joint distribution of (X, Y) are given by

- $p_X(x) = \sum_y p_{X,Y}(x, y)$;
- $p_Y(y) = \sum_x p_{X,Y}(x, y)$

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Independence

Recall that events A and B are independent iff $\mathrm{P}(AB) = \mathrm{P}(A)\mathrm{P}(B)$

Definition (Independent rv)

Discrete **rv** X and Y are independent if for every x and y, the events $A := \{X = x\}$ and $B := \{Y = y\}$ are independent

In terms of **pmf** independence can be expressed as follows:

$$\boxed{\forall x, y : \qquad p_{X,Y}(x, y) = p_X(x)p_Y(y)}$$

Independence

Example (An urn with 3 white and 2 black balls)

	$X=1$	$X=2$	$X=3$	p_Y
$Y=0$	.1	0	0	.1
$Y=1$	.2	.4	0	.6
$Y=2$	0	.2	.1	.3
p_X	.3	.6	.1	

X and Y are dependent as $p_{X,Y}(2,1) \neq p_X(2)p_Y(1)$; in fact, $X \geq Y$

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Conditional distribution

- Given the joint distribution of X and Y and given that $Y = y$ occurred, we can introduce the conditional distribution of the **rv** $X \mid Y = y$:

$$p_{X|Y}(x \mid y) := \mathrm{P}(X = x \mid Y = y) = \frac{\mathrm{P}(X = x, Y = y)}{\mathrm{P}(Y = y)}$$

- For a fixed y, $p_{X|Y}(\cdot \mid y)$ is a **pmf** of the **rv** $X \mid Y = y$:

$$\sum_x p_{X|Y}(x, y) = \sum_x \frac{p_{X,Y}(x, y)}{p_Y(y)} = \frac{p_Y(y)}{p_Y(y)} = 1$$

- If X and Y are independent, then $(X \mid Y = y) \sim X$

Problem

If X and Y are independent Poisson random variables with respective parameters λ_1 and λ_2, calculate the conditional distribution of X given that $X + Y = n$.

Solution

Start with calculating the **pmf** of the **rv** $Z := X + Y$:

$$p_Z(n) = \sum_{k=0}^{n} p_{X,Y}(k, n-k) = \sum_{k=0}^{n} p_X(k) p_Y(n-k)$$

$$= e^{-(\lambda_1+\lambda_2)} \sum_{k=0}^{n} \frac{\lambda_1^k \lambda_2^{n-k}}{k!(n-k)!} = \frac{(\lambda_1+\lambda_2)^n}{n!} e^{-(\lambda_1+\lambda_2)}$$

Thus Z has a Poisson distribution with parameter $\lambda_1 + \lambda_2$, and

$$p_{X|Z}(k \mid n) = \frac{p_{X,Z}(k,n)}{p_Z(n)} = \frac{p_{X,Y}(k, n-k)}{p_Z(n)}$$

$$= \frac{n!}{k!(n-k)!} \frac{\lambda_1^k \lambda_2^{n-k}}{(\lambda_1+\lambda_2)^n} = \binom{n}{k} p^k (1-p)^{n-k}$$

this is a binomial distr. with parameters n and $p = \lambda_1/(\lambda_1+\lambda_2)$